

과정 1 . 5번-7번

The image shows a VS Code editor with two open files. The top window displays `mars_mission_computer.py` with the following code:

```

1 """Mars mission computer that reports sensor readings periodically."""
2
3 import json
4 import random
5 import time
6
7
8 class DummySensor:
9     """A dummy sensor that produces randomized Mars base env values."""
10
11     def __init__(self):
12         self.env_values = {
13             'mars_base_internal_temperature': 0,
14             'mars_base_external_temperature': 0,
15             'mars_base_internal_humidity': 0,
16             'mars_base_external_illumiance': 0,
17             'mars_base_internal_co2': 0,
18             'mars_base_internal_oxygen': 0,
19         }
20
21     def set_env(self):
22         """Fill env_values with random values within their valid range."""
23         self.env_values['mars_base_internal_temperature'] = (
24             random.randint(18, 38))
25         self.env_values['mars_base_external_temperature'] = (
26             random.randint(0, 21))
27         self.env_values['mars_base_internal_humidity'] = (
28             random.randint(50, 68))
29         self.env_values['mars_base_external_illumiance'] = (
30             random.randint(500, 715))
31         self.env_values['mars_base_internal_co2'] = (
32             round(random.uniform(0.82, 0.1), 4))
33         self.env_values['mars_base_internal_oxygen'] = (
34             round(random.uniform(4, 7), 2))
35
36     def get_env(self):
37         """Return the current env_values dictionary."""
38         return self.env_values
39
40
41 """Main Function"""

```

The bottom window displays `main.py` with the following code:

```

1 import numpy as np
2
3 try:
4     arr1 = np.genfromtxt("1-5-mars_base_main_parts-001.csv", delimiter=",",
5                         dtype=str, skip_header=1, encoding="utf-8-sig")
6     arr2 = np.genfromtxt("1-5-mars_base_main_parts-002.csv", delimiter=",",
7                         dtype=str, skip_header=1, encoding="utf-8-sig")
8     arr3 = np.genfromtxt("1-5-mars_base_main_parts-003.csv", delimiter=",",
9                         dtype=str, skip_header=1, encoding="utf-8-sig")
9 except OSError as e:
10     print(f"파일을 읽는 중 오류가 발생했습니다: {e}")
11 else:
12     # 세 파일 모두 부분 이름이 같은 순서로 나열되어 있으므로,
13     # 이름은 arr1 기준으로 사용하고 강도(strength) 값만 합한다.
14     parts_name = arr1[:, 0]
15     strength1 = arr1[:, 1].astype(float)
16     strength2 = arr2[:, 1].astype(float)
17     strength3 = arr3[:, 1].astype(float)
18
19     parts = np.array([strength1, strength2, strength3])
20
21     average = parts.mean(axis=0)
22
23     mask = average < 50
24     weak_parts_name = parts_name[mask]
25     weak_parts_average = average[mask]
26
27     for name, avg in zip(weak_parts_name, weak_parts_average):
28         print(f"{name}: {avg:.2f}")
29
30     try:
31         with open("parts_to_work_on.csv", "w", encoding="utf-8") as f:
32             f.write("parts,average_strength\n")
33             for name, avg in zip(weak_parts_name, weak_parts_average):
34                 f.write(f"{name},{avg:.2f}\n")
35     except Exception as e:
36         print(f"파일 저장 중 오류가 발생했습니다: {e}")
37
38

```

The terminal at the bottom shows the execution output:

```

sej@sejinui-notebug problem6 % python3 main.py
3, 'mars_base_internal_co2': 0.8413, 'mars_base_internal_oxygen': 5.89)
sej@sejinui-notebug problem6 %

```

```
stage1 > problem6 > mars_mission_computer.py > ...
1 """Mars mission computer dummy environment sensor."""
2
3 import random
4
5
6 class DummySensor:
7     """A dummy sensor that produces randomized Mars base env values."""
8
9     def __init__(self):
10         self.env_values = {
11             'mars_base_internal_temperature': 0,
12             'mars_base_external_temperature': 0,
13             'mars_base_internal_humidity': 0,
14             'mars_base_external_illuminance': 0,
15             'mars_base_internal_co2': 0,
16             'mars_base_internal_oxygen': 0,
17         }
18
19     def set_env(self, env_values: dict[str, int]) -> None:
20         """Set the environment values to the given dictionary.
21         (variable) env_values: dict[str, int] their valid range."""
22         self.env_values['mars_base_internal_temperature'] = (
23             random.randint(10, 30))
24         self.env_values['mars_base_external_temperature'] = (
25             random.randint(0, 21))
26         self.env_values['mars_base_internal_humidity'] = (
27             random.randint(50, 60))
28         self.env_values['mars_base_external_illuminance'] = (
29             random.randint(500, 715))
30         self.env_values['mars_base_internal_co2'] = (
31             round(random.uniform(0.02, 0.1), 4))
32         self.env_values['mars_base_internal_oxygen'] = (
33             round(random.uniform(4, 7), 2))
34
35     def get_env(self):
36         """Return the current env_values dictionary."""
37         return self.env_values
38
39 ds = DummySensor()
40 ds.set_env()
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```

PROBLEMS DEBUG CONSOLE PORTS TERMINAL OUTPUT

sejindsejinui-notebug problem6 & python3 main.py
3, 'mars_base_internal_co2': 0.0413, 'mars_base_internal_oxygen': 5.09)
sejindsejinui-notebug problem6 %

Screen Reader Optimized Ln 42, Col 1 Spaces: 4 UTF-8 LF Python Python 3.12.0